Integration Architecture Narrative

Backbone topology, patterns, standards, and the interface estate.

Owner: Liam O'Sullivan · **Source system:** SAP PI/PO · **As of:** 2026-06-06 · **Sensitivity:** Internal

Topology

Lakeshore runs a hybrid integration backbone: SAP PI/PO (ESB) for ERP-centric A2A/B2B, Azure API Management for REST facades to SaaS, Azure Service Bus/Event Grid for event-driven ingestion (including the q-context-ingestion-events queue used by the context loader), and a Managed File Transfer tier for bank host-to-host. See `integration_architecture_diagram.svg` for the component view and interface inventory.

Patterns & standards

Pattern	Where used	Standard/Protocol
Request/response	SaaS lookups via APIM	REST/JSON, OAuth2, mTLS
Async messaging	Context ingestion, eventing	AMQP (Service Bus), pub/sub, DLQ
File-based batch	Bank + ERP bulk	SFTP/MFT, PGP, ISO20022 XML
A2A orchestration	ERP↔TMS↔EPM	IDoc, SOAP, XSLT mapping
CDC/ELT	Source→Snowflake	Snowpipe, log-based CDC

Key end-to-end flows

- Payments: SAP IDoc → Kyriba → ISO20022 pain.001 → bank → pain.002 ack → camt.053 statement → ERP recon.
- Context load: ZIP → Blob (context-drops) → Service Bus → worker → parse → enterprise_context_chunks → AI Search.
- Analytics: SAP/Workday/Coupa → Snowflake medallion → semantic layer → Power BI + AI Search.

Resilience & observability

Guaranteed-once delivery via Service Bus sessions + idempotent sinks; dead-letter queues with replay; interface SLAs monitored in Azure Monitor with incidents routed to ServiceNow. 240+ active interfaces are inventoried with owner, criticality, and recovery objectives.